POLITECNICO DI TORINO

Master's Degree in Electronics

Master's Degree Thesis

Loss analysis of a 4-level,flying capacitor,bridge-less,totem-pole PFC

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Chapter 1

SPICE model Validation

In order to properly analyze more critical components in terms of losses , is useful to first create a digital model of the device,a validation procedure of the SPICE models of each device has been performed. The following sections describes the procedure and results of this validation procedure.

All simulation results are available [1]

1.1 Fast Leg GaN FET model validation

This section describes the behavior of the SPICE model of EPC2304, compared to the EPC datasheet [2]

1.1.1 Static charateristics

Output I-V charateristics

```
.temp 25

.dc Vds 0 6 0.1 Vgs list 2 3 4 5
```

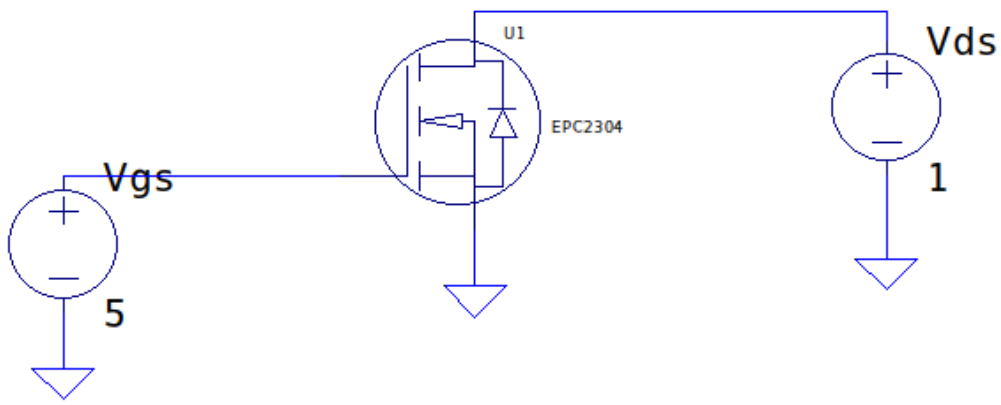

Figure 1.1: Schematic for I-V validation

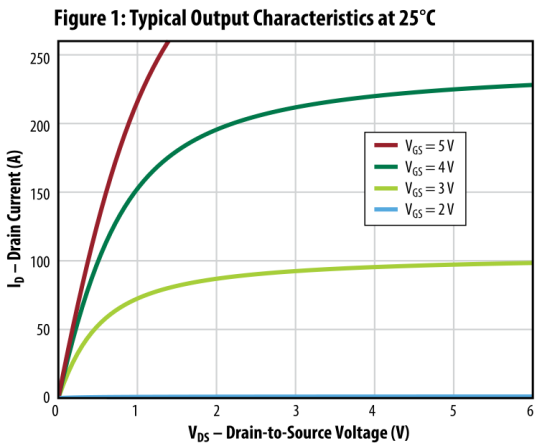

Figure 1.2: Datasheet I-V

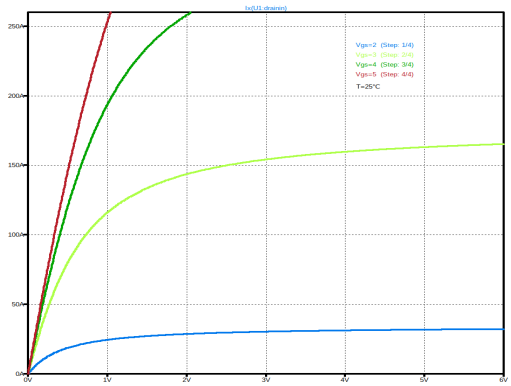

Figure 1.3: SPICE I-V

Vds@Vgs	Ids (datasheet)	Ids (simulation)
Vds=1V;Vgs=2V	$\approx 0A$	25.13A
Vds=6V;Vgs=3V	$\approx 100A$	163.15A
Vds=1V;Vgs=4V	$\approx 150A$	187.12A
Vds=1V;Vgs=5V	$\approx 215A$	253.07A

Table 1.1: I-V data comparision

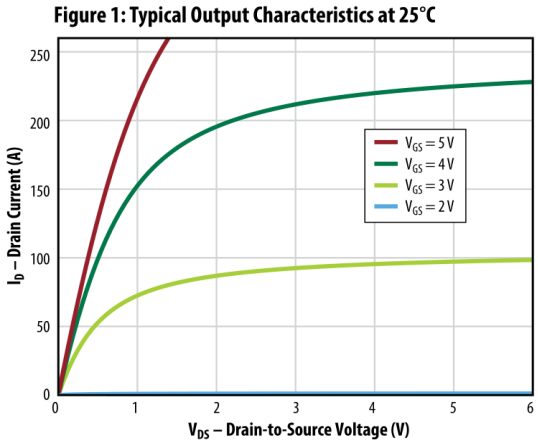

Figure 1.4: Datasheet I-V

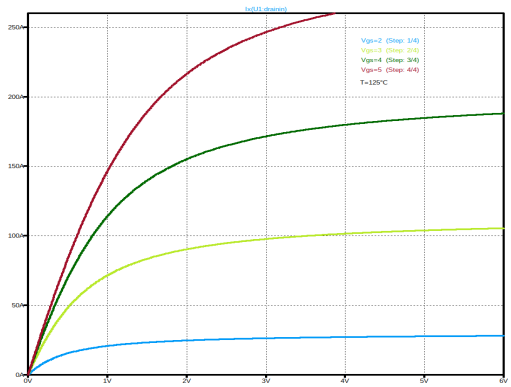

Figure 1.5: SPICE I-V @125°C

Vds@Vgs	Ids (datasheet)	Ids (simulation)
Vds=1V;Vgs=2V	$\approx 0A$	16.41A
Vds=6V;Vgs=3V	$\approx 100A$	105A
Vds=1V;Vgs=4V	$\approx 150A$	110.1A
Vds=1V;Vgs=5V	$\approx 215A$	149.2A

Table 1.2: I-V data comparision

Trans charateristics

```
.step temp 25 125 100

.temp 25

.dc Vgs 0.5 5 0.1
```

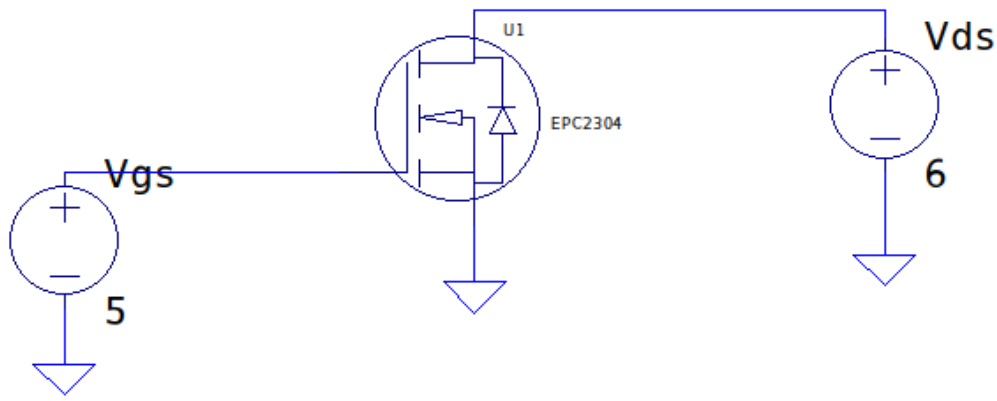

Figure 1.6: Schematic for trans charateristics validation

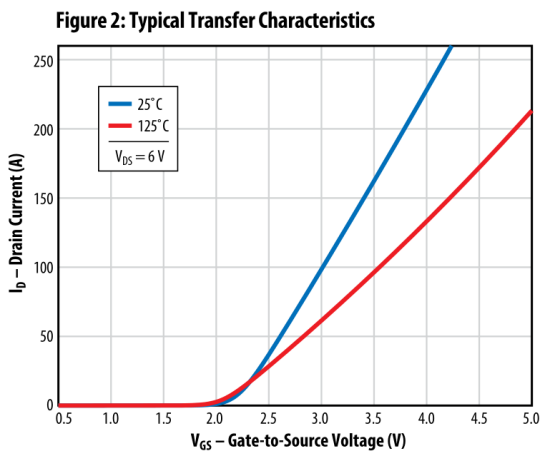

Figure 1.7: Datasheet trans charateristics

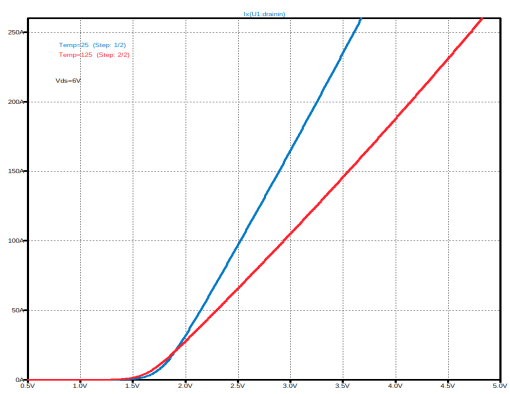

Figure 1.8: SPICE trans charateristics

Comparing the 2 it seems like the real device has a Vth of $\approx 2V$,while the simulated one has a Vth of $\approx 1.5V$

Vgs@T	Ids (datasheet)	Ids (simulation)
Vgs=2.5V;T=25°C	$\approx 40A$	98.58A
Vgs=3V;T=25°C	$\approx 100A$	163.15A
Vgs=3.5;T=125°C	$\approx 100A$	148.12A
Vgs=4.5V;T=125°C	$\approx 175A$	238.07A

Table 1.3: I-V data comparision

Rdson

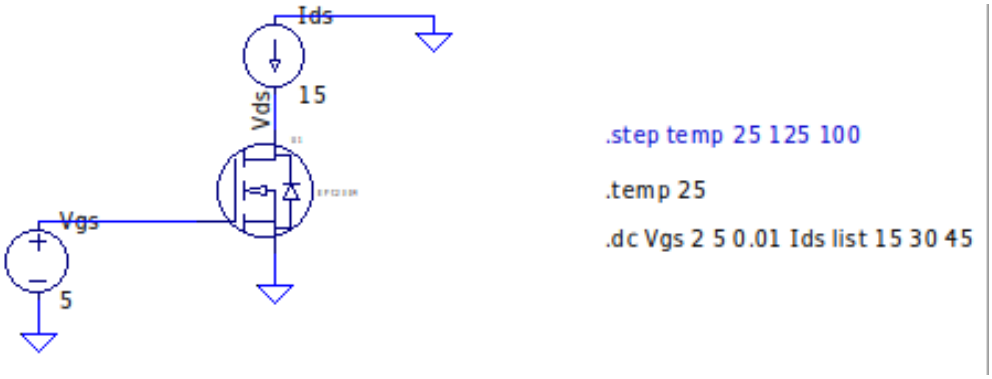

Figure 1.9: Schematic for Rdson validation

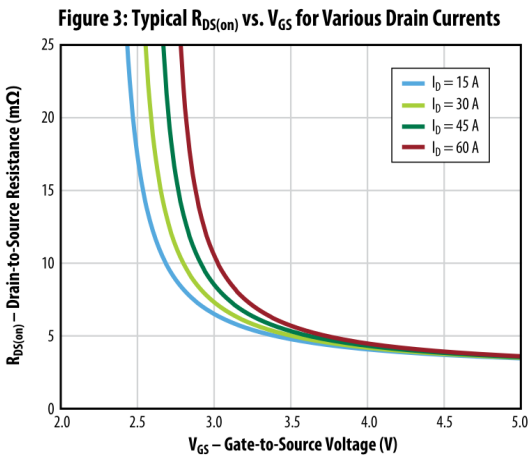

Figure 1.10: Datasheet Rdson

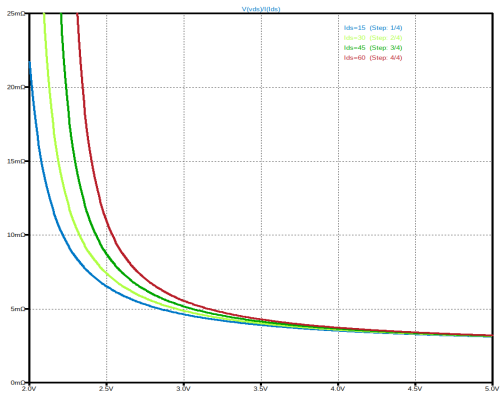

Figure 1.11: SPICE Rdson @25°C

Vgs@Ids	Rds (datasheet)	Rds (simulation)
Vgs=2.5V;Id=15A	$\approx 17.5m\Omega$	$6.8m\Omega$
Vgs=3V;Ids=60A	$\approx 10m\Omega$	$6m\Omega$
Vgs=2V;Ids=15A	$+\infty$	$22m\Omega$
Vgs=5V	$\approx 4m\Omega$	$3.11m\Omega$

Table 1.4: Rds data comparision

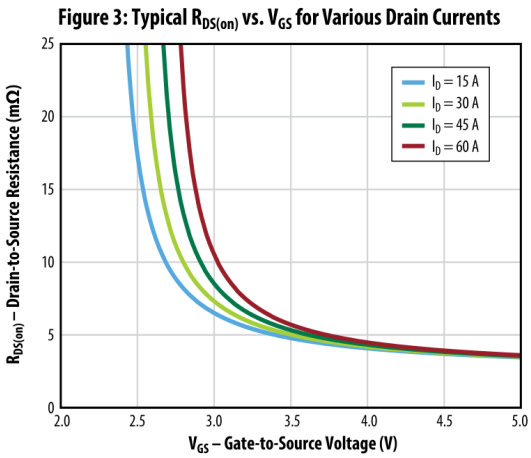

Figure 1.12: Datasheet Rdson

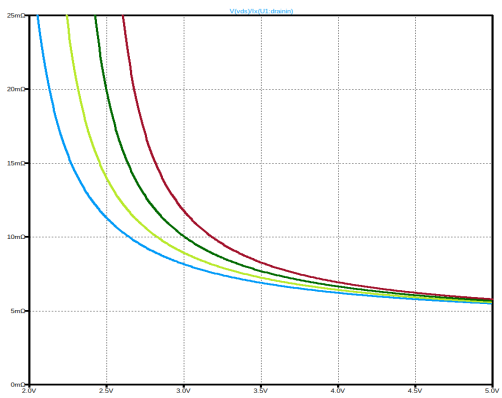

Figure 1.13: SPICE Rdson @125°C

Vgs@Ids	Rds (datasheet)	Rds (simulation)
Vgs=2.5V;Id=15A	$\approx 17.5m\Omega$	$11.8m\Omega$
Vgs=3V;Ids=60A	$\approx 10m\Omega$	$12m\Omega$
Vgs=2V;Ids=15A	$+\infty$	$30.91m\Omega$
Vgs=5V	$\approx 4m\Omega$	$5.49m\Omega$

Table 1.5: Rds data comparision

Rdson vs tempearture

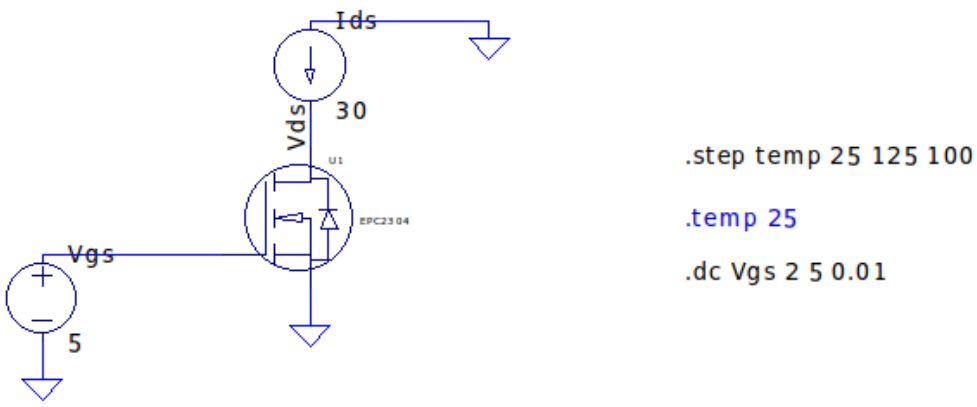

Figure 1.14: Schematic for Rdson vs tempearture

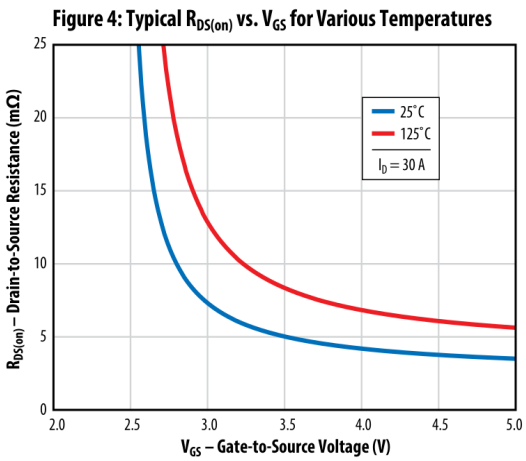

Figure 1.15: Datasheet Rdson vs tempearture

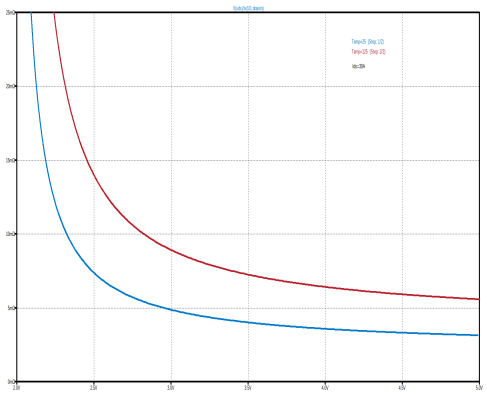

Figure 1.16: SPICE Rdson vs tempearture

Vgs@T	Rds (datasheet)	Rds (simulation)
Vgs=3.5V;T=25°C	$\approx 5m\Omega$	$4.6m\Omega$
Vgs=5V;T=125°C	$\approx 5.5m\Omega$	$5.8m\Omega$
Vgs=2.95V;T=25°C	$\approx 7m\Omega$	$5m\Omega$
Vgs=2.3V@T=25°C	$+\infty$	$10m\Omega$
Vgs=2.3V@T=125°C	$+\infty$	$21.2m\Omega$

Table 1.6: Rds temperature comparision

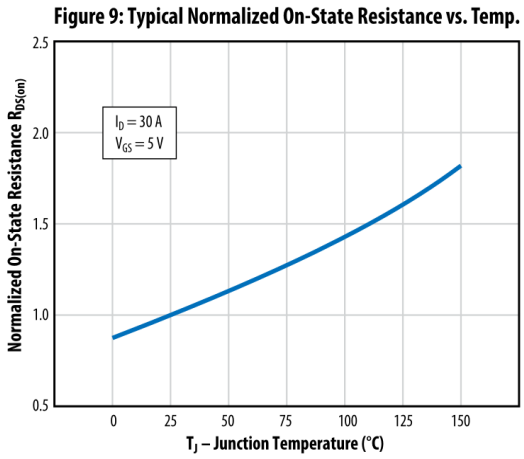

Figure 1.17: Datasheet Rdson vs temperature

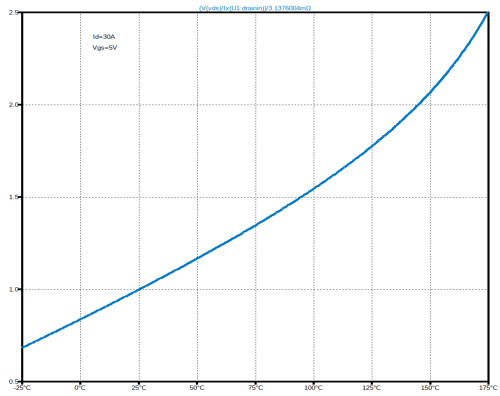

Figure 1.18: SPICE Rdson vs temperature

Reverse body diode

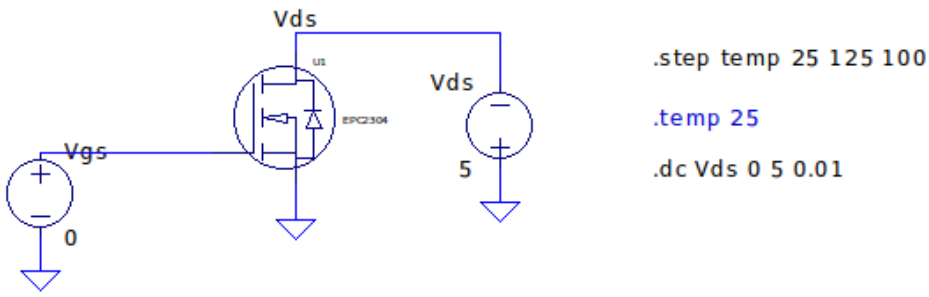

Figure 1.19: Schematic for Reverse body diode

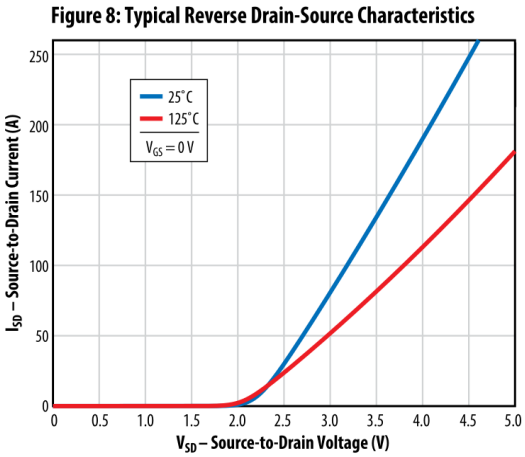

Figure 1.20: Datasheet Reverse body diode

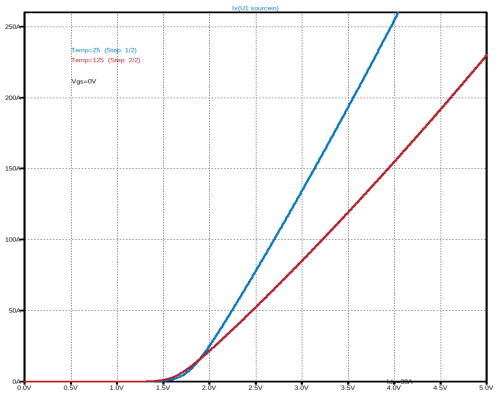

Figure 1.21: SPICE Reverse body diode

Vgs@T	Rds (datasheet)	Rds (simulation)
Vss=4.5V;T=25°C	$\approx 250A$	$317A$
Vsd=3V;T=125°C	$\approx 50A$	$88.7A$
Vsd=4.5V;T=125°C	$\approx 148A$	$191A$

Table 1.7: Reverse diode data comparision

Vth vs temperature

Figure 1.22: Schematic for Vth vs temperature

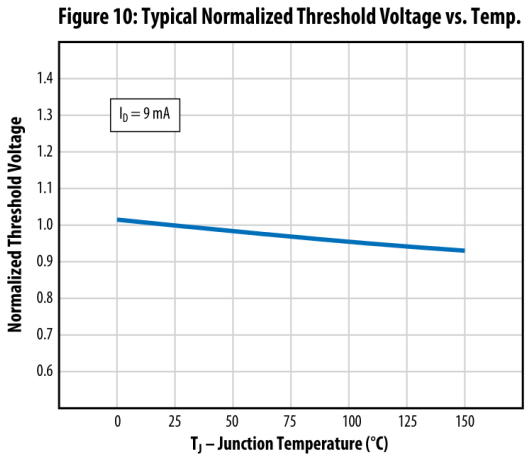

Figure 1.23: Datasheet Vth vs temperature

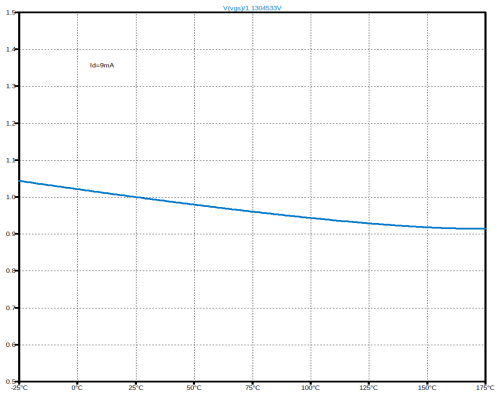

Figure 1.24: SPICE Vth vs temperature

1.1.2 Dynamic Charateristics

Typical switching curves

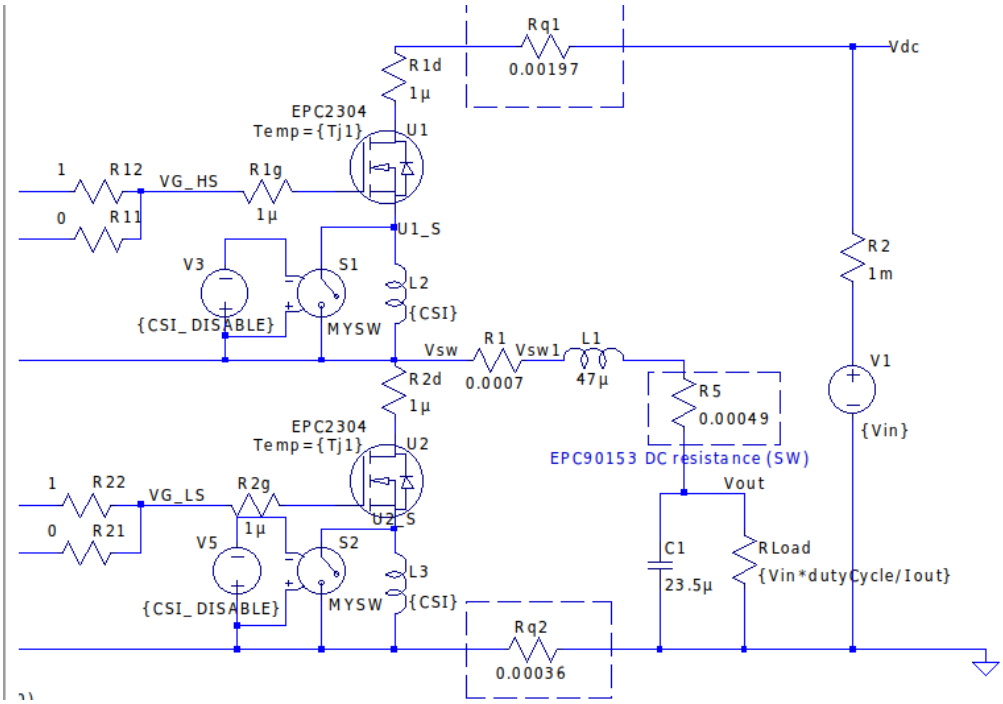

Figure 1.25: Schematic for switching behaviour

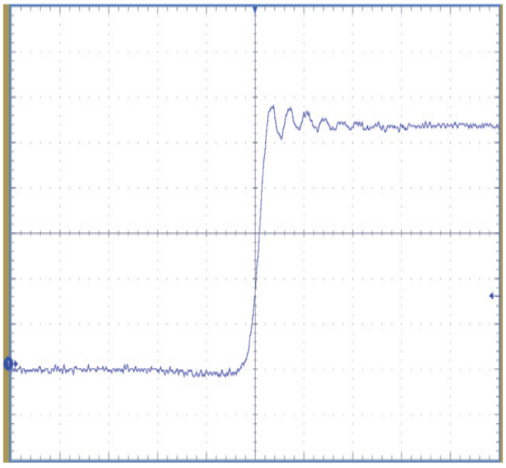

Figure 1.26: Datasheet Turn on curve

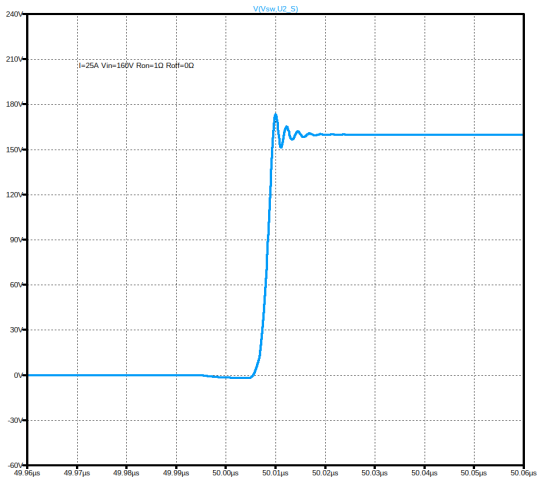

Figure 1.27: SPICE Turn on curve

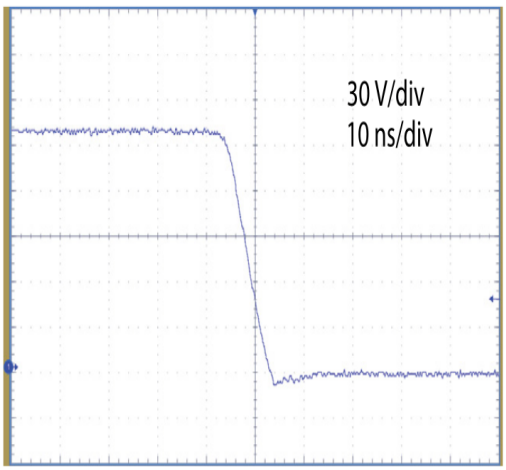

Figure 1.28: Datasheet Turn off curve

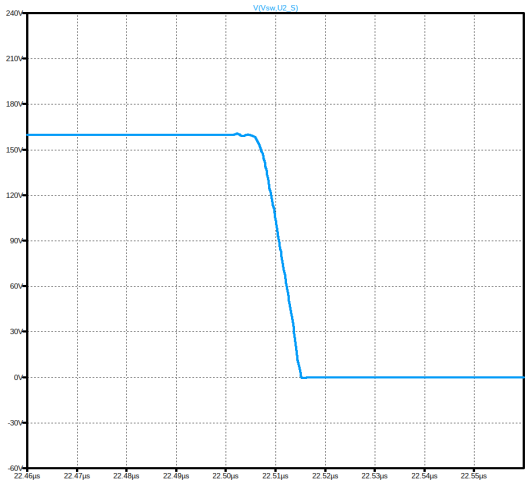

Figure 1.29: SPICE Turn off curve

Typical gate charge

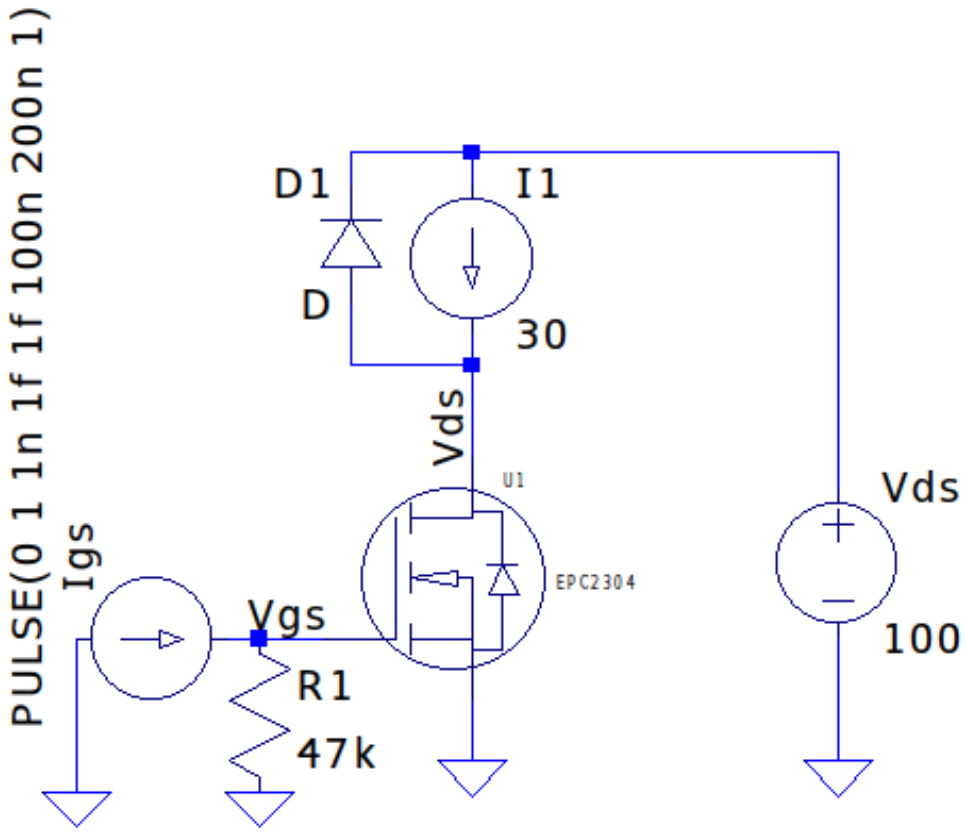

Figure 1.30: Schematic for gate charge

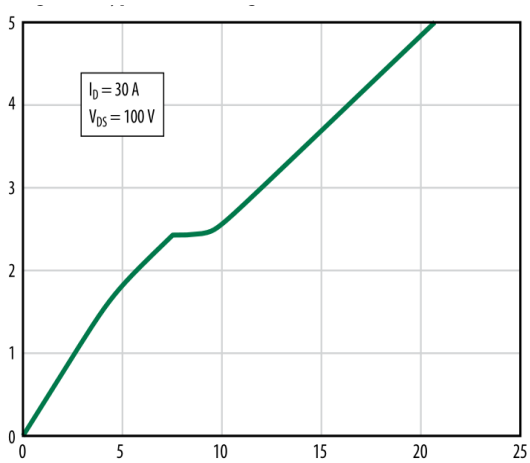

Figure 1.31: Datasheet gate charge

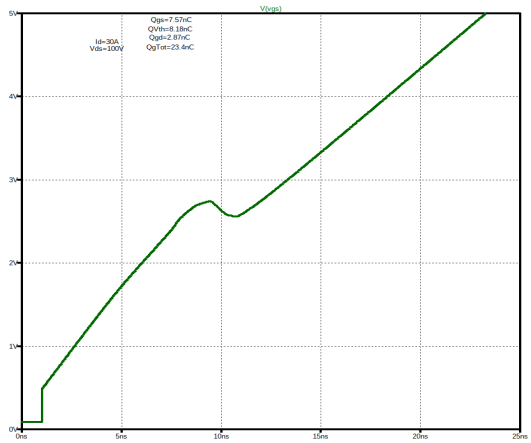

Figure 1.32: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=21nC	Qgtot=23.4nC
Qgs=7.5nC	Qgs=7.57nC
Qgd=2nC	Qgd=2.87nC
QVth=5.2nC	QVth=8.18nC

Table 1.8: Gate charge data comparision

1.2 Slow leg Si FET model validation

This section describes the behavior of the SPICE model of IPZ60R017C7, compared to the datasheet. [3] All results are intended for $T_j = 25°C$ unless otherwise noted.

1.2.1 Static charateristics

Op analysis

Parameter	Datasheet(typ)	Simulation	Notes
$V_{(GS)th}$	3.5V	3.16V	$V_{DS} = V_{GS}$,I_D=2.91mA 1.34
I_{DSS}	$2\mu A$	$3.6\mu A$	$V_{DS} = 600$; $V_{GS} = 0$,$T_j = 25°C$ 1.33
I_{DSS}	$50\mu A$	$3.82\mu A$	$V_{DS} = 600$; $V_{GS} = 0$,$T_j = 125°C$ 1.33
I_{GSS}	$100nA$	$20nA$	$V_{DS} = 0$; $V_{GS} = 20$ 1.33

Table 1.9: static characteristics comparision

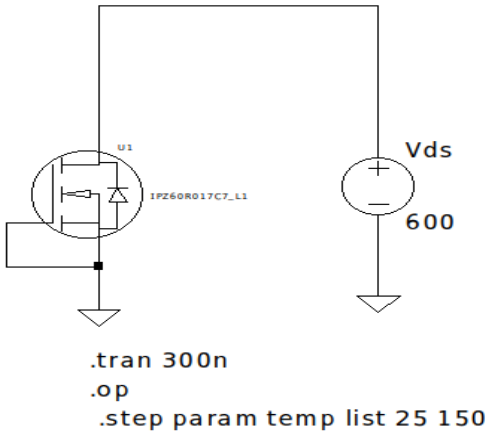

Figure 1.33: zero gate validation circuit

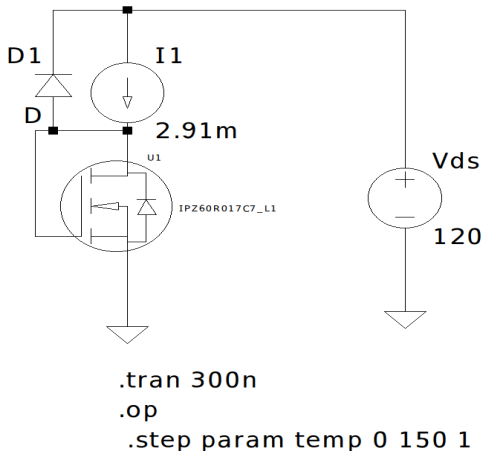

Figure 1.34: Vth validation circuit

Trans charateristics

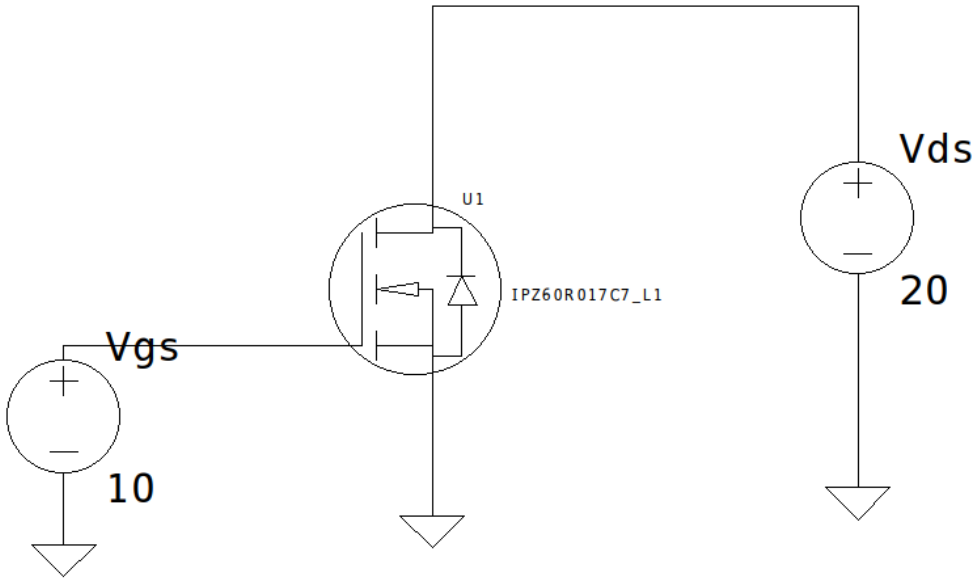

Figure 1.35: Schematic for Trans char validation

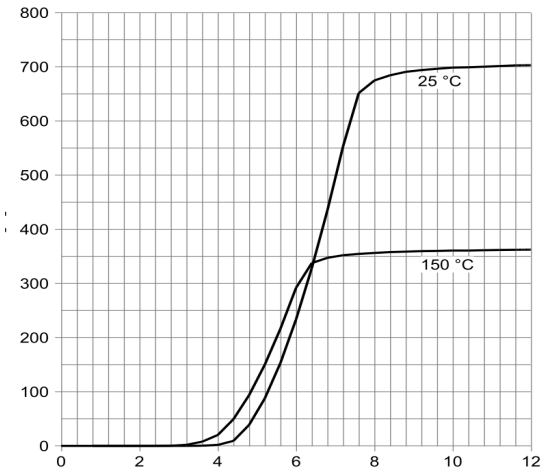

Figure 1.36: Datasheet Trans char

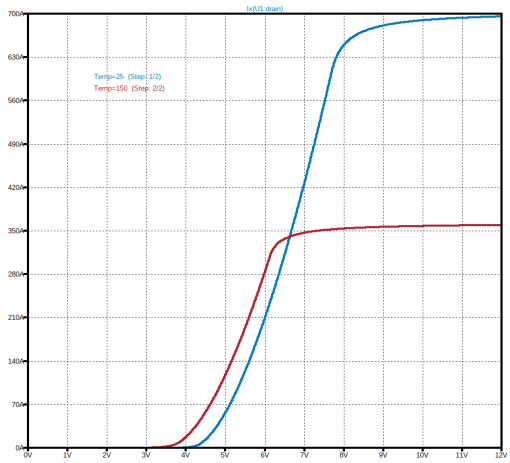

Figure 1.37: SPICE Tran char

T_j;V_{GS}	I_D Datasheet	I_D Simulation	Notes
$T_j = 150°C; V_{GS} = 8V$	$\approx 350A$	353.64	-
$T_j = 25°C; V_{GS} = 10V$	$\approx 700A$	689.57	-
$T_j = 25°C; V_{GS} = 6.2V$	$\approx 340A$	689.57	-
$T_j = 150°C; V_{GS} = 3.2V$	$\approx 0A$	0.533A	Vth
$T_j = 25°C; V_{GS} = 4$	$\approx 0A$	0.684A	Vth

Table 1.10: trans characteristics comparision

Output charateristics

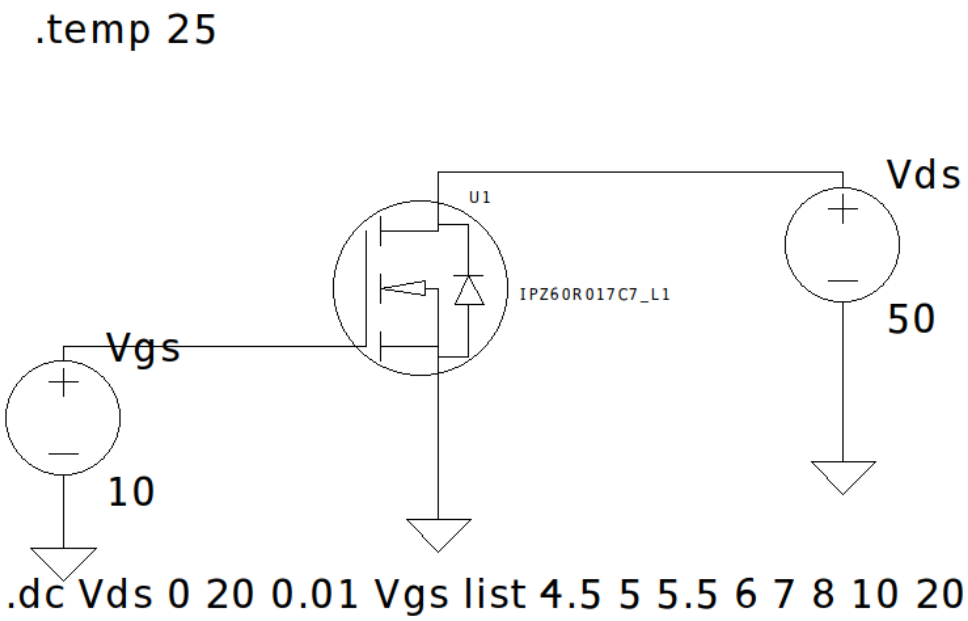

Figure 1.38: Schematic for Si I-V validation

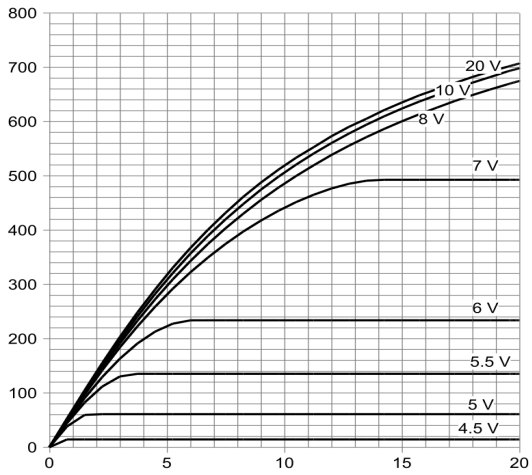

Figure 1.39: Datasheet output char

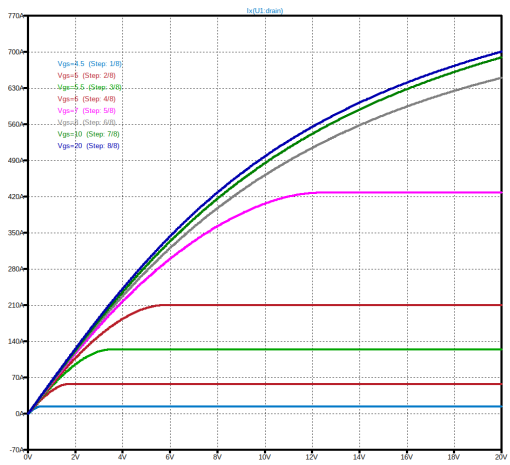

Figure 1.40: SPICE output char

V_{DS};V_{GS}	I_D Datasheet	I_D Simulation
$V_{DS} = 20V$;$V_{GS} = 20V$	$\approx 710A$	701A
$V_{DS} = 20V$;$V_{GS} = 10V$	$\approx 700A$	687A
$V_{DS} = 5V$;$V_{GS} = 4.5V$	$\approx 15A$	13.94A
$V_{DS} = 20V$;$V_{GS} = 10V$	$\approx 700A$	687A
$V_{DS} = 10V$;$V_{GS} = 5V$	$\approx 60A$	57.29A

Table 1.11: output characteristics comparision

Reverse diode charateristics

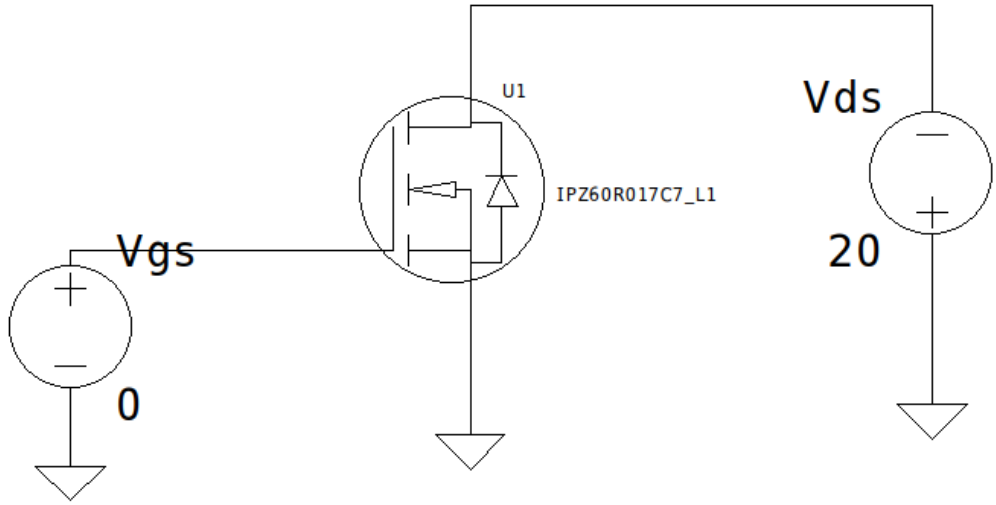

Figure 1.41: Schematic for Reverse diode validation

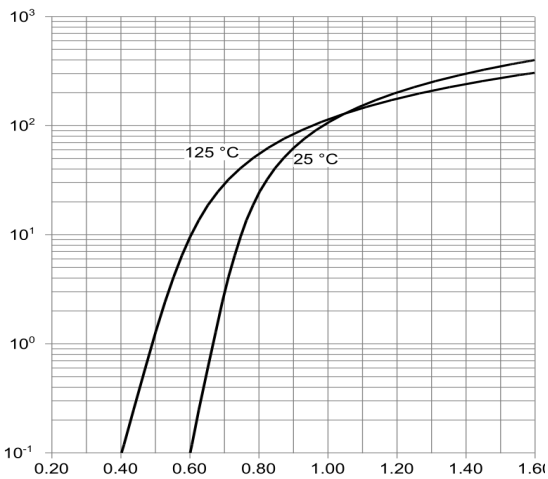

Figure 1.42: Datasheet Reverse diode

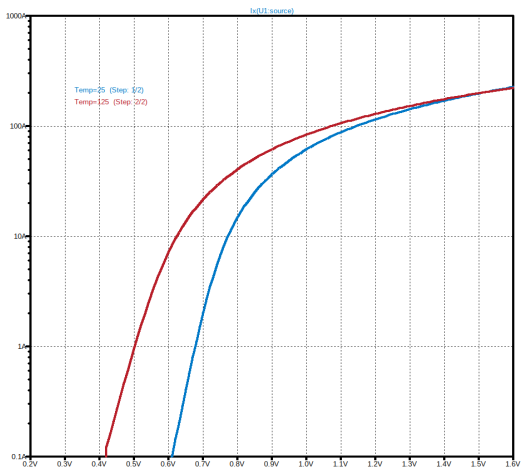

Figure 1.43: SPICE Reverse diode

T_j;V_{DS}	I_D Datasheet	I_D Simulation	Notes
$T_j = 125°C; V_{DS} = 0.4V$	$\approx 100mA$	97.45mA	-
$T_j = 25°C; V_{DS} = 0.6V$	$\approx 100mA$	95.68mA	-
$T_j = 125°C; V_{DS} = 0.5V$	$\approx 1A$	1.03A	-
$T_j = 25°C; V_{DS} = 0.7V$	$\approx 3A$	2.1A	-
$T_j = 25°C; V_{DS} = 0.6V$	$\approx 9.5A$	7.9A	-

Table 1.12: Reverse diode characteristics comparision

1.2.2 Dynamic charateristics

Typical gate charge

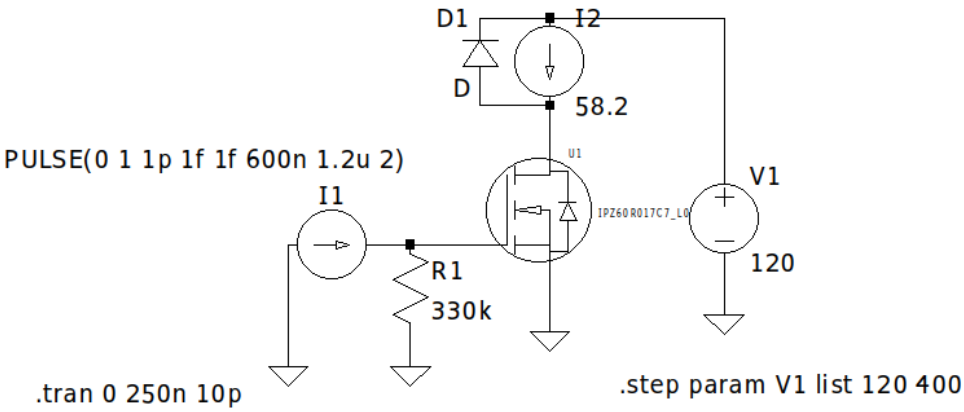

Figure 1.44: Schematic for typical gate charge

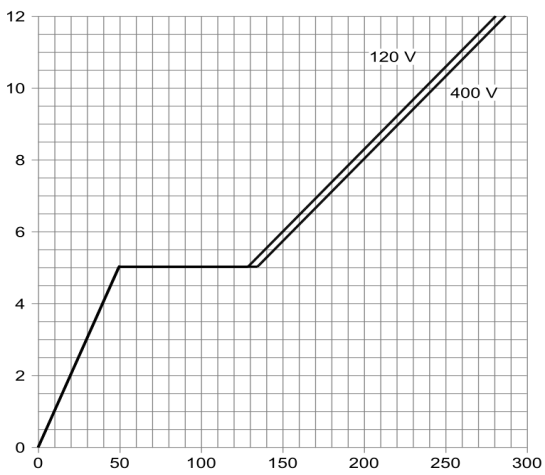

Figure 1.45: Datasheet gate charge

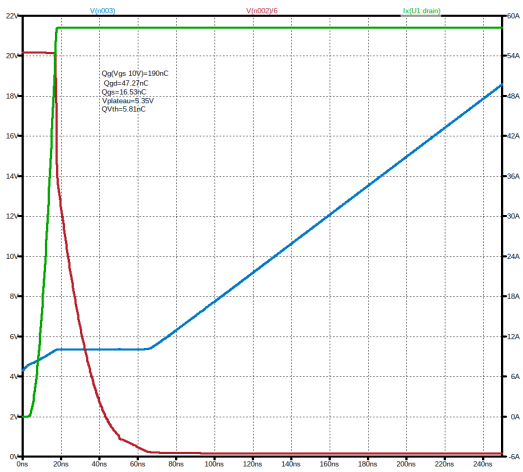

Figure 1.46: SPICE gate charge

Q (datasheet)	Q (simulation)
Qgtot=240nC	Qgtot=190nC
Qgs=50nC	Qgs=16.53nC
Qgd=85nC	Qgd=47.27nC
$V_{V_{th}} = 5V$	$V_{V_{th}} = 5.35V$

Table 1.13: Gate charge data comparision

1.3 Capacitors model validation

This section briefly describes the SPICE model behavior compared to the datasheet described behavior

Flying capacitor

no particular notes needed,model fits real behavior,no temperature model available [4]

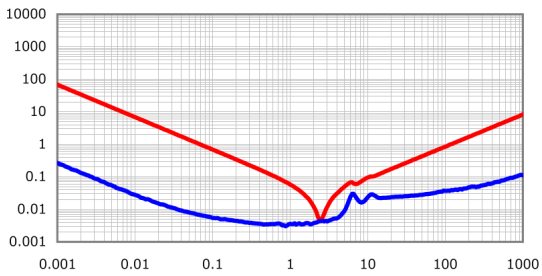

Figure 1.47: Datasheet Z

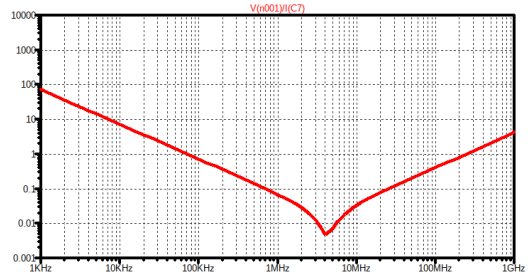

Figure 1.48: SPICE Z

HF decoupling capacitor

no particular notes needed,model fits real behavior,no temperature model available [5]

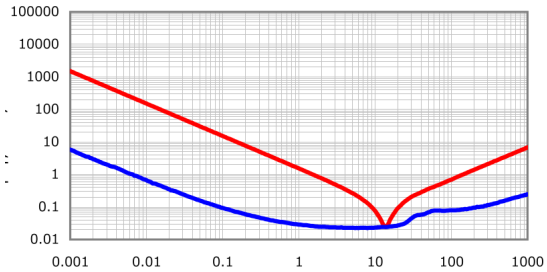

Figure 1.49: Datasheet Z

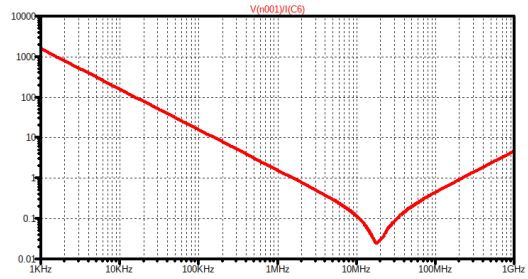

Figure 1.50: SPICE Z

1.4 1EDB7275F Gate driver model validation

This section describes the SPICE model behavior compared to the one described in the datasheet [6] All results are obtained from the circuit shown below and performing a transient analysis,adjusting parameters as needed

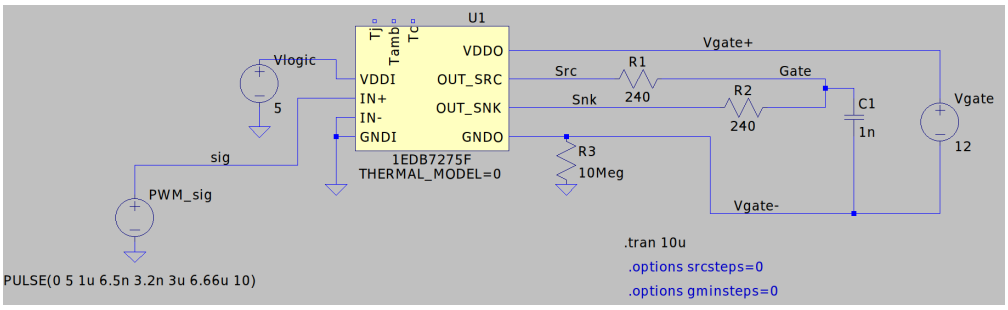

Figure 1.51: Schematic for GaN driver validation

Unless otherwise noted, electrical characteristics are given for VDDI=3.3V, VDDO=12 V Typical values are given at TJ = 25°C

1.4.1 Static characteristics

Parameter	Datasheet(typ)	Simulation	Notes
I_{VDDIq}	$650\mu A$	$654.05\mu A$	OUT=LOW
I_{VDDIq}	$840\mu A$	$670.85\mu A$	OUT=HIGH
V_{INhys}	0.9V	1V	hard coded in model
V_{INH}	2.2V	2.2V	hard coded in model
V_{INL}	1.3V	1.3V	hard coded in model
$I_{IN+,H}$	$40\mu A$	$66.67\mu A$	IN+ tied to VDDI
$I_{IN-,L}$	$-55\mu A$	$71.428\mu A$	IN- tied to GNDI
R_{onSRC}	0.95Ω	1.231Ω	I_{SNK} = 50 mA
R_{onSNK}	0.48Ω	0.55Ω	I_{SRC} = 50 mA

Table 1.14: static characteristics comparision

1.4.2 Dynamic characteristics

Timing characteristics are defined as follows

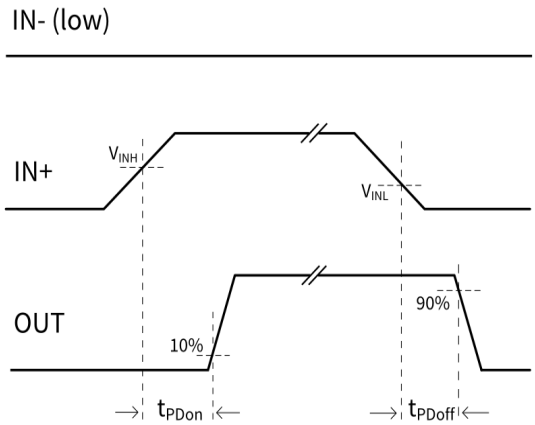

Figure 1.52: Delay definition

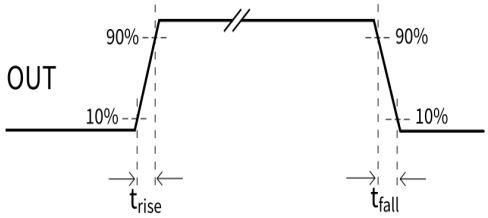

Figure 1.53: Rise/Fall time definitions

Parameter	Datasheet(typ)	Simulation	Notes
I_{SRCpk}	5A	5A	-
I_{SNKpk}	-9A	-8.8A	-
t_{PDon}	45ns	41.39ns	-
t_{PDoff}	45ns	42.66ns	-
t_{rise}	6.5ns	2.30ns	$C_{load} = 1.8nF$
t_{fall}	8ns	9.52ns	$C_{load} = 1.8nF$

Table 1.15: dynamic characteristics comparision

1.5 IRS2183 Half-bridge driver model validation

This section describes the SPICE model behavior compared to the one described in the datasheet [7] All results are obtained from the circuit shown below and performing a transient analysis,adjusting parameters as needed

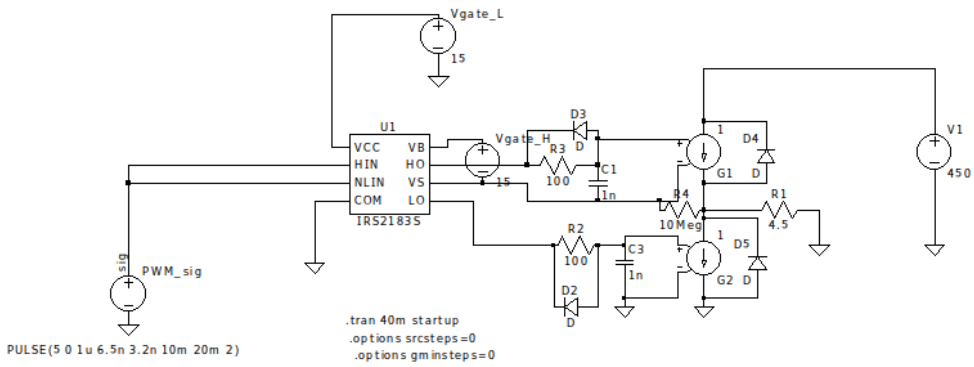

Figure 1.54: Schematic for Si driver validation

Unless otherwise noted, electrical characteristics are given for VCC=VBS=15 V, $C_L = 1nF$,Ta= 25°C

1.5.1 Static characteristics

Parameter	Datasheet(typ)	Simulation	Notes
I_{QBS}	$60\mu A$	$60\mu A$	-
I_{QCC}	1mA	1.026mA	-
I_{IN+}	$25\mu A$	$25\mu A$	-
I_{O+}	1.9A	1.928A	-
I_{O-}	2.31A	1.929A	-

Table 1.16: static characteristics comparision

1.5.2 Dynamic characteristics

Timing characteristics are defined as follows

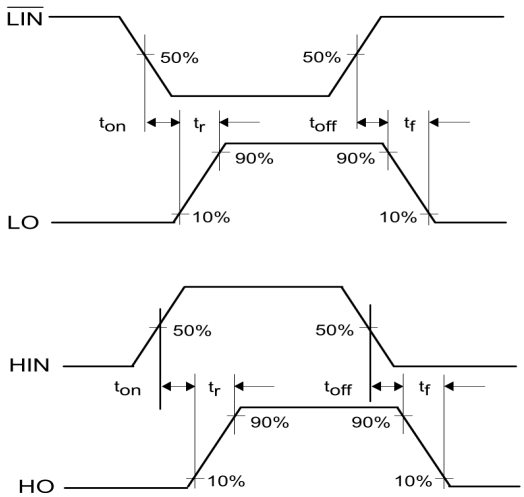

Figure 1.55: Rise,fall,delay definition

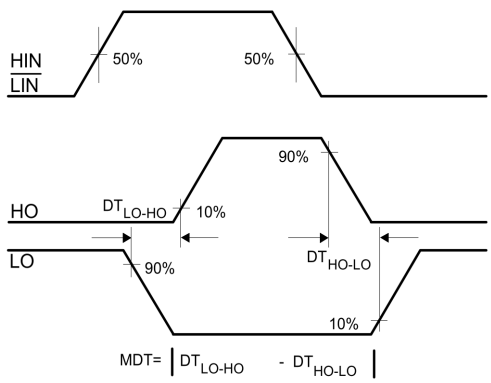

Figure 1.56: Dead time definitions

Parameter	Datasheet(typ)	Simulation	Notes
t_{on}	180ns	186.37ns	-
t_{off}	220ns	190.15ns	-
MT	0ns	4ns	-
t_r	40ns	51.33ns	-
t_f	20ns	22ns	-
DT	400ns	433ns	-

Table 1.17: dynamic characteristics comparision

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